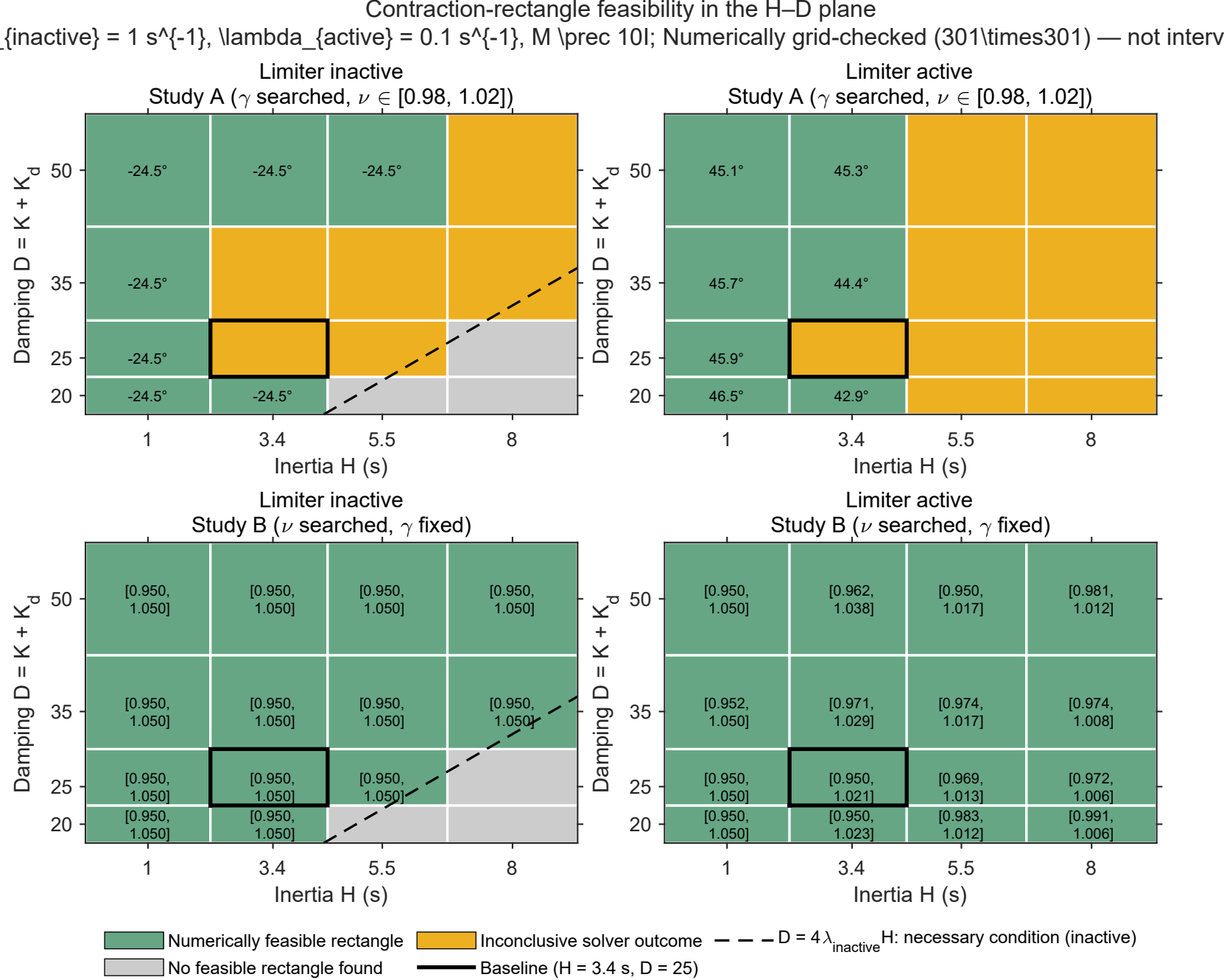
Contraction-rectangle feasibility in the H–D plane
{inactive} = 1 s^{-1}, \lambda_{active} = 0.1 s^{-1}, M \prec 10I; Numerically grid-checked (301\times301) — not interv
Limiter inactive
Study A (γ searched, $\nu \in [0.98, 1.02]$)
Limiter active
Study A (γ searched, $\nu \in [0.98, 1.02]$)
Limiter inactive
Study B (ν searched, γ fixed)
Limiter active
Study B (ν searched, γ fixed)
Damping D = K + K_d
Inertia H (s)
50
35
25
20
1
3.4
5.5
8
-24.5°
45.1°
45.3°
45.7°
44.4°
45.9°
46.5°
42.9°
[0.950, 1.050]
[0.962, 1.038]
[0.950, 1.017]
[0.981, 1.012]
[0.952, 1.050]
[0.971, 1.029]
[0.974, 1.017]
[0.974, 1.008]
[0.950, 1.021]
[0.969, 1.013]
[0.972, 1.006]
[0.983, 1.012]
[0.991, 1.006]
[0.950, 1.023]
Numerically feasible rectangle
Inconclusive solver outcome
D = 4 λ_inactive H: necessary condition (inactive)
No feasible rectangle found
Baseline (H = 3.4 s, D = 25)